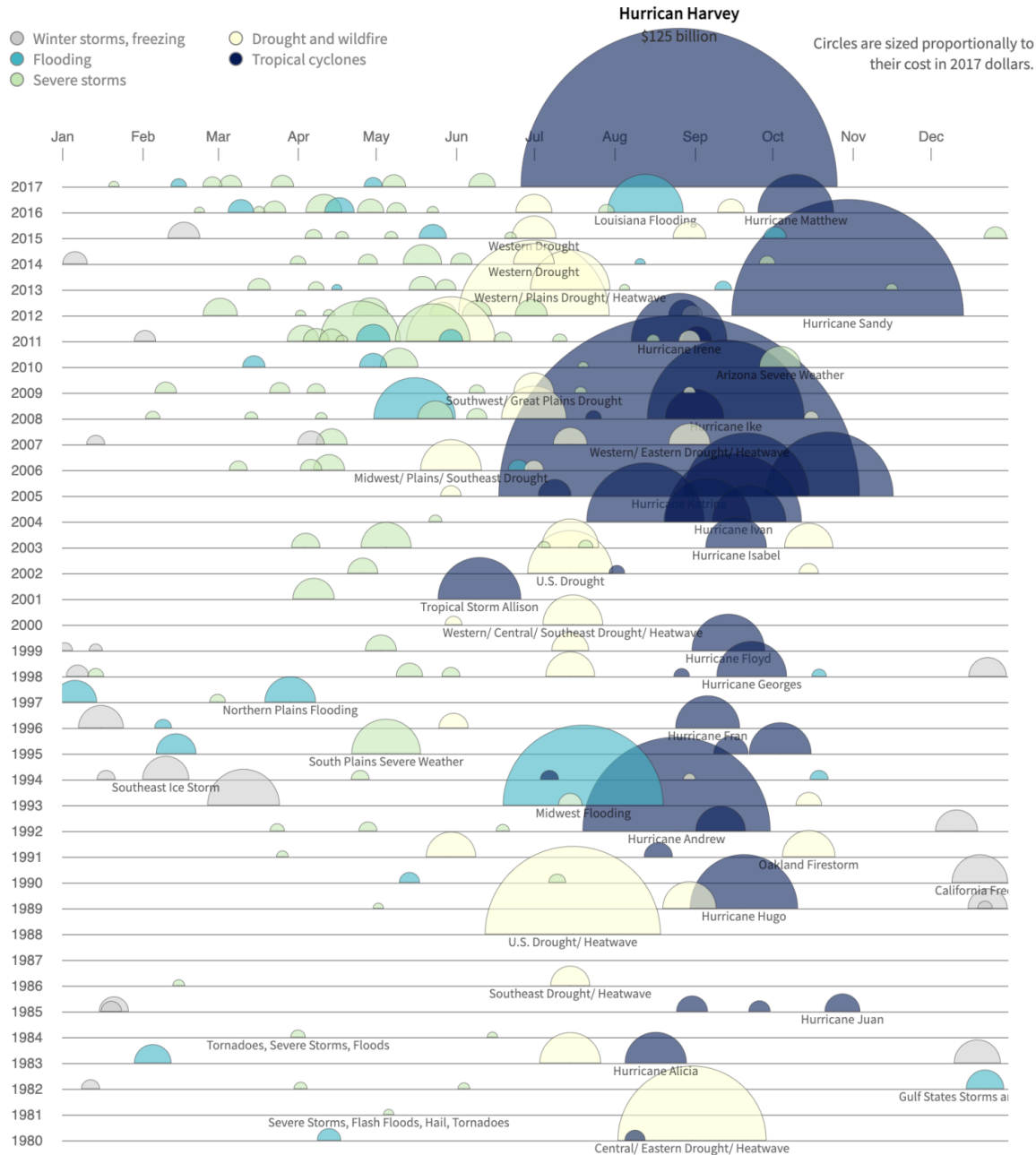


Today's activity: We are going to use the example we have been working on, and will add some interactions.

The Cost of Natural Disasters



This case study is a reproduction of the New York Times' visualization in [The Cost of Hurricane Harvey: Only One Recent Storm Comes Close](#) by Kevin Quealy.

Instructions:

1. Download the template you are going to fill in:

https://drive.google.com/file/d/1VpwNgIBk6wGfF4bmQomqr9Pb76WJpZH/view?usp=drive_link

2. Go to the folder containing the visualization application, and run a simple server. You'll need python.

4. Open the files with a simple text editor or visual studio code.

5. Look over the code. We are going to add:

- Tool tips for the events
- Buttons to filter out events by disaster type

6. Tool-tips step by step:

- You can reference this example:
- First you need to create a tooltip div in your html. I did this for you, check the index.html file
- Then you need to add some style rules for your tooltip. I went ahead and did this for you in the Activity8 template, so check it out (style/style.css).
- TO DO: When you append your circles to the svg (when you create the event circles) you need to add an event listener for mouseover and mouseleave events. Below is the example from the trails scatterplot code. Can you adapt this example to the disasters data- show the event name, the event cost, the date. Look at the datafile to see the options. This is using template literals, which are between `` symbols. You can insert custom data into this string literal using `\${ what to insert }`.

```
.on('mouseover', (event,d) => {
  d3.select('#tooltip')
    .style('display', 'block')
    .style('left', (event.pageX + vis.config.tooltipPadding) + 'px')
    .style('top', (event.pageY + vis.config.tooltipPadding) + 'px')
    .html(`
      <div class="tooltip-title">${d.name}</div>
      <div><i>${d.region}</i></div>
      <ul>
        <li>${d.distance} km, ~${d.time} hours</li>
        <li>${d.difficulty}</li>
        <li>${d.season}</li>
      </ul>
    `);
})
```

```
.on('mouseleave', () => {  
    d3.select('#tooltip').style('display', 'none');  
});
```

7. Filtering step by step

- First, I added buttons. Look at them in the html file
- Next, I added style rules for the buttons, go look in the style sheet. These will opacity when selected/deselected, and they have stored the name of the disaster type, so you can know what the user selected/deselected.
- Then you have to handle the click event. Look in main. Everything should be there. Walk through it, step-by-step.
- TO DO: Then, go to the timeline circles file. Look over how some of the code is now in 'update'. We are updating which events are shown. Something needs to change. What should change?

If you have time- examine the style rules for the buttons. Try some other settings, to see if you can make hovering. selection and deselection clearer.